Grade 6 Accelerated
Unit 1
Lesson 3 Homework

Name______________________
Date______________________
Period______________________

Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
1. 37×9	$9(30+7)$ $= 270 + 63$ $= 333$	__30__ + __7__ __9__ | 270 | 63	333
2. 8×54	$8(50+4)$ $= 400 + 32$ $= 432$	__50__ + __4__ __8__ | 400 | 32	432
3. 95×6	$6(90+5)$ $= 540 + 30$ $= 570$	__90__ + __5__ __6__ | 540 | 30	570

For each expression, generate an equivalent numeric expression using the distributive property and then find the value.

4. 7×23
$7(20+3)$
$= 7 \times 20 + 7 \times 3$
$= 140 + 21$
$= 161$

5. $15 \cdot 9$
$9(10+5)$
$= 9 \times 10 + 9 \times 5$
$= 90 + 45$
$= 135$

6. 5×48
$5(40+8)$
$= 5 \times 40 + 5 \times 8$
$= 200 + 40$
$= 240$

Rewrite an equivalent expression for each sum using the distributive property with the greatest common factor (GCF).

7. $63 + 105$

$21(3 + 5)$

8. $270 + 414$

$18(15 + 23)$

9. Generate two different equivalent expressions for the sum $168 + 216$ by rewriting it using the distributive property with two different common factors.

Answers may vary.

$12(14 + 18)$
$24(7 + 9)$

10. Zoriah was using the distributive property to write an expression equivalent to $108 + 16$. Her expression was $4(27 + 16)$. Is Zoriah correct? If yes, explain how she determined her expression. If not, explain Zoriah's error and write a correct equivalent expression.

Her expression is incorrect. $4 \times 27 = 108$ but $4 \times 16 = 64$ not 16. If she wanted to use the common factor of 4, the correct expression would be $4(27 + 4)$.